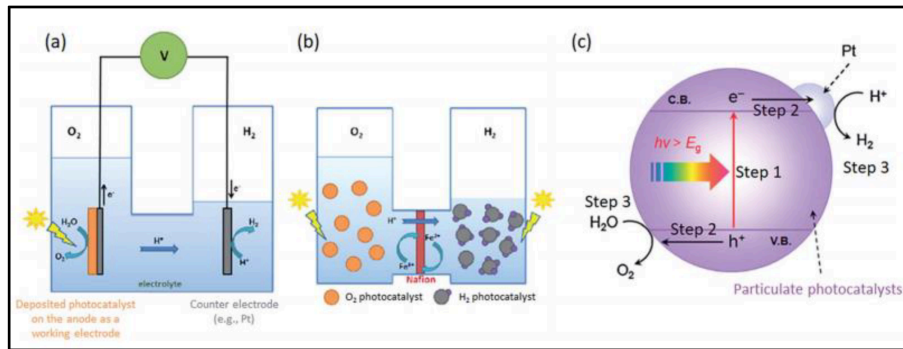


Application of semiconductor in nanodevices

Semiconductor nanomaterials have interesting physical and chemical properties and useful functionalities. Narrow and intensive emission spectra, continuous absorption bands, high chemical and photobleaching stability, processability, and surface functionality are among the most attractive properties of these materials. Semiconductor nanomaterials have been the focus of research for about 20 years and have attracted significant interest in research and applications in diverse disciplines such as solid-state physics, inorganic chemistry, physical chemistry, colloid chemistry, materials science, and recently biological sciences, medical sciences, engineering, and interdisciplinary fields. Among the unique properties of nanomaterials, the movement of electrons and holes in semiconductor nanomaterials is primarily governed by the well-known quantum confinement, and the transport properties related to phonons and photons are largely affected by the size and geometry of the materials. These novel properties of semiconductor nanomaterials have attracted significant attention in research and applications in emerging technologies such as nanoelectronics, nanophotonics, energy conversion, nonlinear optics, miniaturized sensors and imaging devices, solar cells, catalysis, detectors, photography, biomedicine etc., In this section we discuss the application of semiconductor nanomaterials in catalysis and medical sciences.

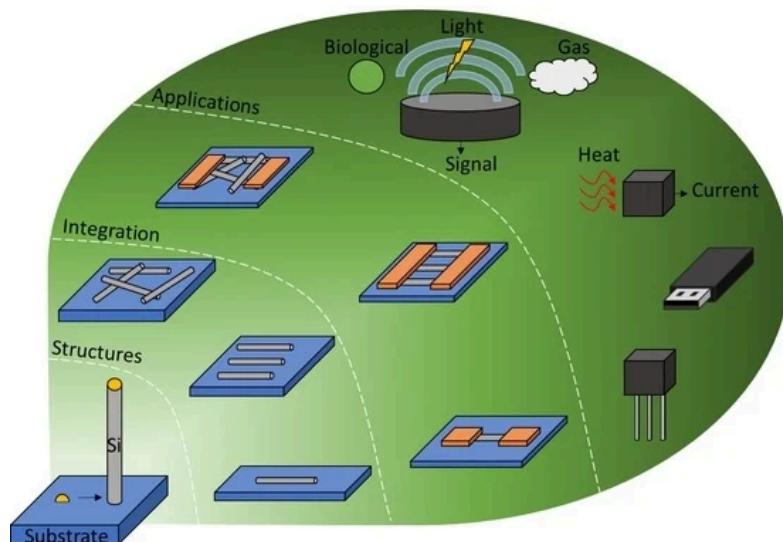
1. Semiconductor Nanomaterials for Hydrogen Production:

The extensive use of fossil fuel over the last 150 years has caused a rise in urban ill-health, economic dependence, political unrest and many cases of warfare[46]. A recent European study revealed that more deaths are caused by car emissions than by car accidents[47], and a parallel Swedish study reported that pollution can increase the risk of having cancer by 40-70 %[48] and recent atmospheric investigations have also shown a substantial increase in the concentrations of carbon dioxide and other greenhouse gases, with potentially threatening consequences to the global climate. Growing environmental concerns are related to the extensive use of non-sustainable fossil fuels (oil, natural gas and coal) and a constantly increasing energy demand for clean and sustainable sources of energy. Hydrogen is a promising alternative fuel, since it is completely pollution-free and can readily be produced from renewable energy resources. thus eliminating the net production of greenhouse gases. Recent studies have indicated that hydrogen fuel costs are reasonable and hydrogen is therefore an ideal candidate to replace fossil fuels as an energy carrier. The production of hydrogen from liquid hydrocarbon fuels and adsorption on various solids including carbon has attracted a lot of attention by the media and the automotive industry. Photocatalysis is expected to make a great contribution to both environmental treatment (emission cleaning and water purification) and renewable energy. Hydrogen (H₂) is widely considered to be the future clean energy carrier in many applications, such as environmentally friendly vehicles, domestic heating, and stationary power generation. Photocatalytic



2. Silicon Semiconductor Nanomaterials and Devices

This field has become one of the most active research areas of semiconductor nanotechnology. Knowledge of modern information technology is based on silicon based microelectronics. Therefore, the development of silicon based materials and devices will influence the future development of current microelectronics and information technology to a certain extent, which has an important strategic significance. Besides the widely used crystalline silicon wafers, silicon nanostructures have been considered as the basic components for future nano/microelectronic devices. With the size of silicon based complementary metal oxide semiconductors (CMOS) circuit reducing gradually down to 10 nm or even smaller, silicon based nanotechnology can provide a solution with lower cost and higher efficiency. The size effect of silicon nanostructures leads to a lot of novel properties. By using these novel properties scientists have recently developed several silicon based novel devices like highly sensitive biological and chemical sensors, high-efficiency solar cells and light-emitting diodes.



3.Organic Optoelectronic Materials and Devices

organic optoelectronic materials with strong photoluminescence, nonlinear optical materials, and some important low molecular weight pharmaceutical materials, construct the nanodevices and explore their practical applications. Organic optoelectronics has developed into a new interdisciplinary research field, involving organic chemistry, physics, electronic engineering and materials science. Organic optoelectronic devices, such as organic electroluminescent devices (OLED), organic photovoltaic (OPV) and organic thin film transistors (OTFT), have attracted significant attention in academics and industries, due to their great application potential in flat-panel and flexible display, solid-state lighting, information transport and storage, new generation energy, photocatalyst and so on. Owing to the advantages of solid-state, self-emission, full color capability and flexibility, OLED has been recognized as one of the most promising flat-panel display technologies and has just become commercial.

